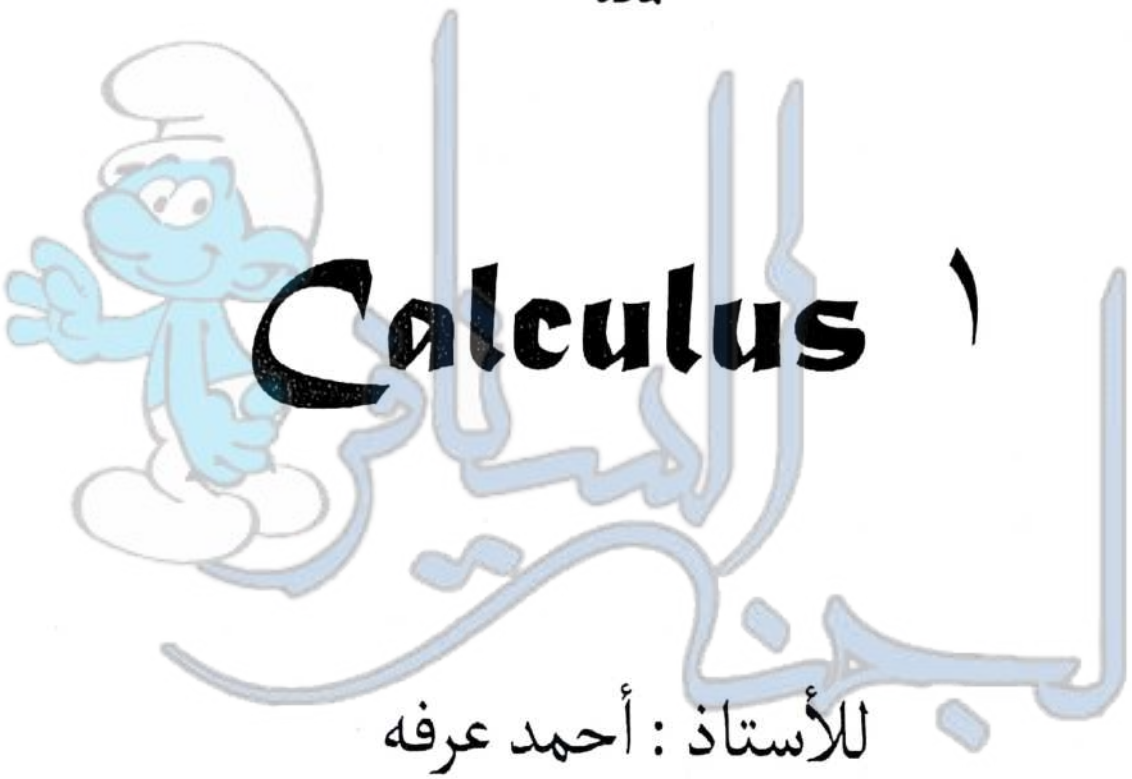
# حلول أسئلة السنوات

٢٠١٧/ ٢٠١٨

لمادة

# Calculus ١




للأستاذ : أحمد عرفه

٠٧٨٦٥٤٣٦٦٥


كل سؤال يحوي اقتران أسي $e^x$ أو لوغارتمي Ln x /Log x احتفظ به لكالكولس ٢ .


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Calculus 1

Midterm "2017/2018-1st semester"

Q1) The range of the function $f(x) = \frac{1}{\sqrt{4-x^2}}$ is

$a)(0,2)$ $b)[0,2]$ $c)(\frac{1}{2},\infty)$ $d)[\frac{1}{2},\infty)$

Q2) Given that $f(x) = \frac{1}{x-3}$ and $g(x) = \frac{1}{x}$ then the domain of the function $f \circ g$ is

$a)\mathbb{R} \setminus \{0, \frac{1}{3}\}$ $b)\mathbb{R} \setminus \{0\}$ $c)\mathbb{R} \setminus \{\frac{1}{3}\}$ $d)\mathbb{R} \setminus \{0,3\}$

Q3)( The greatest integer less than or equals x) The range of $f(x) = 2[x]$ is

$a)\mathbb{R}\setminus\{0, \pm1, \pm2, \pm3, .....\}$ $b)\{0, \pm1, \pm2, \pm3, .....\}$ $c)(-\infty,\infty)$ $d)\{0, \pm2, \pm4, \pm6, .....\}$

Q4) If the domain of the function $y = f(x)$ is $[2,3)$ then the domain of $g(x) = f(3-x)$ is

$a)[2,3)$ $b)(0,1]$ $c)[0,1)$ $d)(2,3]$

Q5) Given that $f(x) = x^2$ and $g(x) = \begin{cases} 2x & x < 4 \\ x+1 & x \geq 4 \end{cases}$ then $(f \circ g)(x) =$

$a)$ $\begin{cases} 4x^2 & x < 4 \\ (x+1)^2 & x \geq 4 \end{cases}$

$b)$ $\begin{cases} 4x^2 & x \leq 4 \\ (x+1)^2 & x > 4 \end{cases}$

$c)$ $\begin{cases} 4x^2 & x < 16 \\ (x+1)^2 & x \geq 16 \end{cases}$

$d)$ $\begin{cases} 4x^2 & x \leq 16 \\ (x+1)^2 & x > 16 \end{cases}$

Q6) The domain of the function $f(x) = \ln(5 - |7x+3|)$ is

$a)[-\frac{8}{7}, \frac{2}{7}]$ $b)\mathbb{R} \setminus [-\frac{8}{7}, \frac{2}{7}]$ $c)(-\frac{8}{7}, \frac{2}{7})$ $d)\mathbb{R} \setminus (-\frac{8}{7}, \frac{2}{7})$

Q7) The $x$-intercept of the function $f(x) = -2 + \ln(3x+1)$ is

$a)0$ $b)\frac{1}{3}e^2$ $c)\frac{1}{3}(e^2-1)$ $d)\frac{1}{3}(1-e^2)$



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Q8) The solution set of the inequality $e^{2x} > 5e^{x}$ is

a)$(-\infty, \ln\sqrt{5})$ b)$(\ln\sqrt{5}, \infty)$ c)$(0, \frac{1}{2}\ln 5)$ d)$(-\frac{1}{2}\ln 5, 0)$

Q9) The domain of $f(x) = \cos^{-1}(3x+1)$ is

a)$[-1, 1]$ b)$[-\frac{1}{2}, 0]$ c)$[-\frac{2}{3}, 0]$ d)$[0, \frac{2}{3}]$

Q10) The range of the function $f(x) = 4\sin^{-1}(x^2)$ is

a)$[-\frac{\pi}{2}, \frac{\pi}{2}]$ b)$[-2\pi, 2\pi]$ c)$[0, 2\pi]$ d)$[-2\pi, 0]$

Q11) Given that $f(x) = \csc^{-1} x$ then $f(2) =$

a)$\frac{1}{\sin 2}$ b)$\frac{1}{\cos 2}$ c)$\cos^{-1}(\frac{1}{2})$ d)$\sin^{-1}(\frac{1}{2})$

Q12) If the $f(x) = x^2 - 4x + 7,,\ x \in [k, \infty)$ is one to one function , then the smallest possible value of $k$ is

a)4 b)2 c)$-2$ d)$-4$

Q13) Given that $f(x) = \pi + \cot^{-1}(x-2)$then $f(2) =$

a)$\frac{\pi}{2}$ b)$\frac{3}{2}\pi$ c)$\pi$ d)$undefined$

Q14) The value of $4\sin^{-1}(\frac{\sqrt{5}}{19}) + 4\cos^{-1}(\frac{\sqrt{5}}{19}) =$

a)$\pi$ b)$\frac{1}{2}\pi$ c)$\frac{1}{4}\pi$ d)$2\pi$

Q15) $\lim_{x\to-\infty}(\frac{\sqrt{49x^2+x+1}+2x+9}{6x+4}) =$

a)$\frac{-5}{6}$ b)$\frac{3}{2}$ c)$\infty$ d)$-\infty$



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Q16) $\lim_{x\to 2^-}(\frac{1}{x-2}-\frac{12}{x^3-8})=$

a)1 b) $-1$ c) $\frac{-1}{2}$ d) $\frac{1}{2}$

Q17) $\lim_{x\to 4^+}(\frac{x^2-6x+8}{\sqrt{x^2-8x+16}}=$

a)2 b) $-2$ c)0 d)*does not exist*

Q18) If $y=0$ and $x=2$ are horizontal nd vertical asymptotes (respectively) of the function $f(x)=\frac{ax^3+x(3x+1)}{x^3-b}$ then $a+b=$

a)11 b)8 c) $-1$ d)5

Q19)The function $f(x)=x^2\ln|x|$ is

a)*even* b)*odd* c)*neither odd nor even* d)*positive for all* $x\in$ *Domain*$(f)$

Q20) $f(x)=\begin{cases}2-x & x\le 0\\ 1+e^{-x} & x\ge 0\end{cases}$ then $f^{-1}(x)=$

a) $\begin{cases}x-2 & x>2\\ -\ln(x-1) & 1<x\le 2\end{cases}$

b) $\begin{cases}2-x & x>2\\ -\ln(x-1) & 1<x\le 2\end{cases}$

c) $\begin{cases}2-x & x>2\\ -1+\ln x & 1<x\le 2\end{cases}$

d) $\begin{cases}x-2 & x>2\\ -1+\ln x & 1<x\le 2\end{cases}$

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*Good Luck*



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Q1:-

Sol:- $x^2 \geq 0$

$-x^2 \leq 0$

$4 - x^2 \leq 4$

$\sqrt{4-x^2} \leq 2$

$\frac{1}{\sqrt{4-x^2}} \geq \frac{1}{2}$

$[\frac{1}{2}, \infty)$ (d)

Q2:- $f(x) = \frac{1}{x-3}$, $g(x) = \frac{1}{x}$

$D_{f \circ g} = ?$

Sol: $(f \circ g)(x) = f(g(x))$

$= f(\frac{1}{x}) = \frac{1}{\frac{1}{x} - 3}$

$\frac{1}{x} - 3 = 0 \Rightarrow \frac{1}{x} = 3$

$\Rightarrow x = \frac{1}{3}$

$D_{f(g(x))} = R - \{\frac{1}{3}\}$.

$D_{f \circ g} = D_{f(g(x))} \cap D(g(x))$

$= (R - \{\frac{1}{3}\}) \cap (R^+ \cup \{0\})$

$= R - \{0, \frac{1}{3}\}$ (a)

Q3:- Range $[x] = \{0, \pm 1, \pm 2, \pm 3, \ldots\}$

Rang $2[x] = \{0, \pm 2, \pm 4, \pm 6 \ldots\}$

(d)

Q4: $Df(x) = [2, 3)$ then

$Df(3-x)$ is ?

Sol: $2 \leq 3 - x < 3$

$(-3) \quad (-3) \quad (-3)$

$\Rightarrow -1 \leq -x < 0$

$1 \geq x > 0 \Rightarrow (0, 1]$ (b)

Q5:-

$g(x) = 2x \rightarrow f(2x) = (2x)^2 = 4x^2$

$g(x) = x + 1 \rightarrow f(x+1) = (x+1)^2$

$(f \circ g)(x) = f(g(x))$

$= \begin{cases} (2x)^2, & x < 4 \\ (x+1)^2, & x \geq 4 \end{cases}$

$= \begin{cases} 4x^2 & x < 4 \\ (x+1)^2 & x \geq 4 \end{cases}$

(a)

Q6:- حذ وف

Q7:-

The x-intercept → y=0

$0 = -2 + \ln(3x+1)$

$2 = \ln(3x+1)$

$e^2 = 3x+1$

$e^2 - 1 = 3x$

$x = \frac{e^2-1}{3}$ (C)

Q8: حذوف

Q9:- $-1 \le 3x+1 \le 1$

$-2 \le 3x \le 0$

$\frac{-2}{3} \le x \le 0$

$[\frac{-2}{3}, 0]$

Q10:- Range $\sin^{-1}(x^2) = [\frac{-\pi}{2}, \frac{\pi}{2}]$

Range $4\sin^{-1}(x^2) = 4[\frac{-\pi}{2}, \frac{\pi}{2}]$

$= [-2\pi; 2\pi]$

(B)

Q11:- $f(x) = \csc^{-1} x$, then $f(2) =$

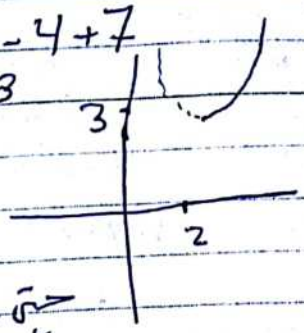
Sol: $\csc^{-1} 2 = \sin^{-1}(\frac{1}{2}) = \frac{\pi}{6}$

(D)

Q12: $f(x) = x^2 - 4x + 7$, $x \in [k, \infty)$ is 1-1

Sol $x^2 - 4x + 4 - 4 + 7$

$(x-2)^2 + 3$




$k = 2$

حتى يكون 1-1 يجب أن يكون مجاله $[2, \infty)$

Q13: $f(x) = \pi + \cot^{-1}(x-2)$ then $f(2) =$

$f(2) = \pi + \cot^{-1}(0)$

$= \pi + \frac{\pi}{2} = \frac{3\pi}{2}$ (B)

Q14: Rule $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

Sol $4\sin^{-1}(\frac{\sqrt{5}}{19}) + 4\cos^{-1}(\frac{\sqrt{5}}{19})$

$= 4(\sin^{-1}(\frac{\sqrt{5}}{19}) + \cos^{-1}(\frac{\sqrt{5}}{19}))$

$= 4(\frac{\pi}{2}) = 2\pi$ (d)

Q15: $\lim_{x \to -\infty} \frac{\sqrt{49x^2 + x + 1} + 2x + 9}{6x + 4} =$

$= \lim_{x \to -\infty} \frac{\sqrt{49}\,|x| + 2x}{6x}$

نأخذ أكبر قوة بالبسط مع أكبر قوة بالمقام مع مراعاة أن بالبسط جزء لها نفس القوة $\sqrt{49x^2}$ ، $2x$

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$\sqrt{x^2} = |x|$

$x \to \infty$ أي إن

$|x| \to -x$ فإن

$$\lim_{x\to-\infty} \frac{-7x+2x}{6x} = \lim_{x\to-\infty} \frac{-5x}{6x}$$

$$= \frac{-5}{6}$$

---

Q16 :

$$\lim_{x\to 2^-} \left(\frac{1}{x-2} - \frac{12}{x^3-8}\right)$$

Sol $x^3-8 = (x-2)(x^2+2x+4)$

$$\lim_{x\to 2^-} \frac{x^2+2x+4-12}{x^3-8}$$

$$= \lim_{x\to 2^-} \frac{x^2+2x-8}{x^3-8}$$

$$= \lim_{x\to 2^-} \frac{(x-2)(x+4)}{(x-2)(x^2+2x+4)}$$

$$= \frac{6}{12} = \frac{1}{2}$$

---

Q17 :

$$\lim_{x\to 4^+} \frac{x^2-6x+8}{\sqrt{(x-4)^2}}$$

Sol

$$\lim_{x\to 4^+} \frac{(x-4)(x-2)}{|x-4|}$$

$|x-4|$ : $-$ | $+$ at $4$

$$= \lim_{x\to 4^+} \frac{(x-4)(x-2)}{x-4} = 4-2$$

$= 2$ (a)

---

Q18 : $y=0$ and $x=2$.

Sol $y=0$

$$\lim_{x\to\pm\infty} \frac{ax^3+3x^2+x}{x^3-b} = 0$$

$$\lim_{x\to\pm\infty} \frac{a}{1} = 0$$

$a=0$

$x=2$

$$\Rightarrow \lim_{x\to 2} \frac{x(3x+1)}{x^3-b} = \pm\infty$$

$$\frac{(2)(7)}{8-b} = \frac{14}{8-b} = \pm\infty$$

if $b=8$

$\therefore a+b = 0+8 = 8$ (b)

---

Q19 : $f(-x) = (-x)^2 \ln|x|$

$= x^2 \ln x$

$= f(x)$

$\therefore f$ is even (a)

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Q20 ⇒

$y = 2 - x$

$x = 2 - y$

$\Rightarrow y = 2 - x$

$f^{-1}(x) = 2 - x$

$y = 1 + e^{-x}$

$x = 1 + e^{-y}$

$x - 1 = e^{-y}$

$\ln(x-1) = \ln e^{-y}$

$\ln(x-1) = -y$

$y = -\ln(x-1)$

$f^{-1}(x) = -\ln(x-1)$

$$f^{-1}(x) = \begin{cases} 2 - x & , \ x > 2 \\ -\ln(x-1) & , \ 1 < x \leq 2 \end{cases}$$

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Balqa Al-salt 2015

Q1) Fill The Following Blanks :

1- If $f(3x) = 9x + 5$, then $f(x) =$

2- The graph $f(x) = x \sin x$ is symmetric around

3- Is the function $f(x) = \frac{x\sqrt{|x|}}{x^4-3}$ even, odd or neither

4- The graph of $\sqrt{x+2} - 3$ is obtained from the graph of $\sqrt{x}$ by and

5- The graph of $-x^2+8x-16$ is obtained from the graph of $x^2$ by and

6- If $f(x) = \sqrt{x-1}$, $g(x) = \frac{1}{x^2-5}$, then the $D_{g \circ f}$ is

7- If $f(x) = 2x^3 - 3x^2 + 1$, and $f^{-1}(x) = 1$, then $x =$

8- $\log_x(2x+3) = 2$, then $x =$

9-The domain of $f(x) = 1 + \sqrt{16 - x^2}$ is

10- If the domain and range of $f(x)$ are $[-1, 3)$ and $(0, 4]$, respectively, then the domain of $g(x) = 1 - 3f(3 - 2x)$ is

11- The range of $f(x) = \sin^{-1}(3x + 1) + \frac{\pi}{2}$ is

12- $\sin^{-1}(\sin \frac{6\pi}{7})$

Q2) If $f(x) = \frac{e^x}{1-5e^x}$ . Find

a) Domain $(f)$ b) $f^{-1}$

c) Range $f$

Good Luck

Ahmad Arafeh

2015

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Q1)

1- $y = 3x$

$\rightarrow x = \frac{y}{3}$

$f(y) = 9\left(\frac{y}{3}\right) + 5$

$f(y) = 3y + 5$

$\therefore f(x) = 3x + 5$

2- around y-axis

$f(-x) = -x \sin(-x)$  نفرض مكان كل $x \rightarrow -x$

Since $\sin x$ is odd.

$= -x \cdot -\sin(x)$

$= x \cdot \sin x$

3- $f(x)$ is odd

$f(-x) = \frac{(-x)^5 |-x|}{(-x)^4 - 3} = \frac{-x^5 |x|}{x^4 - 3}$

$-f(x) = \frac{-x^5 |x|}{x^4 - 3}$

$\therefore f(-x) = -f(x)$

4- Two Units to the left and three Units to the down.

$\sqrt{x+2} - 3$

يسار/يمين ← $a$ →  خارجي أو $b$

5- $-(x^2 - 8x - 16)$

$-(x^2 - 8x + 16 - 16 - 16)$  اكمال مربع

$-((x-4)^2 - 32)$

$32 - (x-4)^2$

4 Units to the right and 32 Units upward and reflection about x-axis.

6- Dgof

$gof(x) = g(f(x)) = g(\sqrt{x-1})$

$= \frac{1}{x-1-5} = \frac{1}{x-6}$

$D_{gof} = D_{\text{مركب}} \cap D_f$

$= (\mathbb{R} - \{6\}) \cap [1, \infty)$

$= [1, \infty) - \{6\}$

7-

$x = f(1)$

$= 2(1) - 3(1) + 1$

$x = 0$

8-

$x^2 = 2x + 3$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1)$

$x = 3 \quad , \quad x = -1$

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9-

$16-x^2 \geq 0$

$x^2 \leq 16$

$\sqrt{x^2} \leq \sqrt{16}$

$|x| \leq 4$

$-4 \leq x \leq 4$

$Df(x) = [-4,4]$

10-

$-1 \leq 3-2x \leq 3$

$-3 \quad -3 \quad -3$

$-4 \leq -2x \leq 0$

عند القسمة على سالب تعكس المتباينة

$\frac{-4}{-2} \geqslant \frac{-2x}{-2} \geqslant \frac{0}{-2}$

$0 \leq x \leq 2$  $Dg(x) = [0,2]$

11- $R_{\sin^{-1}x} = \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$R_{\sin^{-1}(3x+1)+\frac{\pi}{2}} = \left[\frac{-\pi}{2}, \frac{\pi}{2}\right] + \frac{\pi}{2}$

هذا ما يؤثر

$= [0, \pi]$

12-

$\sin^{-1}\left(\sin\frac{6\pi}{7}\right)$

$= \sin^{-1}\left(\sin\frac{\pi}{7}\right) = \frac{\pi}{7}$

Since $\frac{\pi}{7} \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

Q2) a) $Df =$

أصفار المقام

$1-5e^x = 0$

$5e^x = 1$

$e^x = \frac{1}{5}$

$x = \ln\left(\frac{1}{5}\right)$

$Df = (R \cap R) - \left\{\ln\left(\frac{1}{5}\right)\right\}$

$= R - \left\{\ln\left(\frac{1}{5}\right)\right\}$

b) $y = \frac{e^x}{1-5e^x}$

Sol. $x = \frac{e^y}{1-5e^y}$

$x - 5xe^y = e^y$

$x = e^y + 5xe^y$

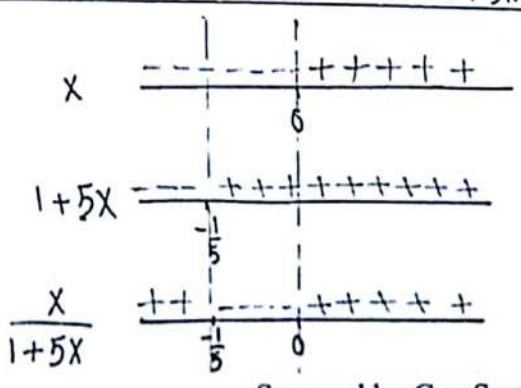
$x = e^y(1+5x)$

$e^y = \frac{x}{1+5x} \Rightarrow y = \ln\left(\frac{x}{1+5x}\right)$

$\therefore f^{-1}(x) = \ln\left(\frac{x}{1+5x}\right)$

c) $R_f = D_{f^{-1}}$

$= \left(-\infty, -\frac{1}{5}\right), (0, \infty)$

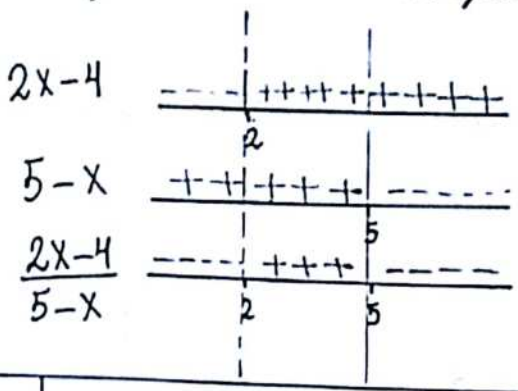



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Q1)

1-




$D f(x) = [2, 5)$

2-

$-1 \le \cos x \le 1$

$-1 \le -\cos x \le 1$

$4-1 \le 4-\cos x \le 4+1$

$3 \le 4-\cos x \le 5$

$\frac{1}{5} \le \frac{1}{4-\cos x} \le \frac{1}{3}$

$\frac{3}{5} \le \frac{3}{4-\cos x} \le 1$

$R f(x) = [\frac{3}{5}, 1]$

3-

$f(-x) = \frac{\sin(-x)}{(-x)^3-(-x)^2} = \frac{-\sin x}{-x^3-x^2} \neq f(x)$

$-f(x) = \frac{-\sin(x)}{x^3-x^2} \neq f(-x)$

∴ neither.

4- about y-axis

$y^3 = 2(-x)^4 - 5(-x)^6$

$y^3 = 2x^4 - 5x^6$

5- $f(g(x)) = (|x|+7)^3$ (التعويض)

$g(x) = |x|+7$

$f(x) = x^3$

6-

$2 \le 3x+1 \le 5$

$-1 \quad -1 \quad -1$

$1 \le 3x \le 4$

$\frac{1}{3} \le x \le \frac{4}{3}$

$D g(x) = [\frac{1}{3}, \frac{4}{3}]$

7- اشتغل على هذا

$x^2-2x+2 \xrightarrow{\text{بدي}} x^2-4x+3$

نكمل المربع

$(x-1)^2+1 \quad (x-2)^2-1$

كيف يتغير ليصبح هكذا

One Unit to the left and two Units upward.

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8- $y = \frac{3x-4}{2-x}$

$x = \frac{3y-4}{2-y}$

$2x - xy = 3y - 4$

$2x + 4 = 3y + xy$

$2x + 4 = y(3 + x)$

$y = \frac{2x+4}{3+x}$

$\therefore f^{-1}(x) = \frac{2x+4}{3+x}$

9- y- axis

10- $gof(1) = g(f(1))$

$= g(-3)$

g is even

$= g(3)$

$= 6$

f odd

$f(-1) = 3$

$-f(1) = 3$

$f(1) = -3$

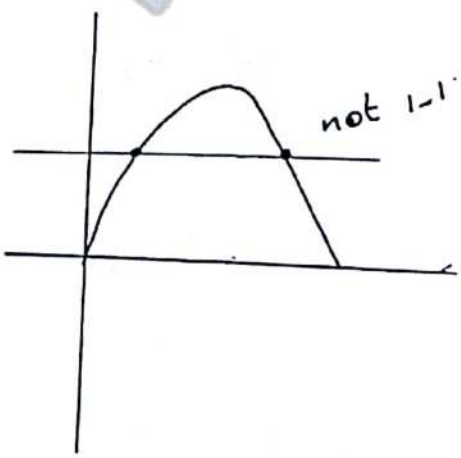
11- $fog(x) = f(g(x))$

$= f(\sqrt{3x-6}) = \frac{1}{2-\sqrt{3x-6}}$

Q2) [d] $\sin x$ , $0 < x < \pi$

$[\frac{-\pi}{2}, \frac{\pi}{2}]$ هو $\sin x$ لأن مجال

حتى يكون له انعكاس




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*Mid – Exam* 21 – 3 – 2017

Q1) Evaluate the limit

$$\lim_{x\to 2} \frac{1}{4x^2 - 16}$$

Q2) Find the value(s) of $x$ (if any) that satisfy

$$\sin(3x) = 0, \quad x \in [0, \frac{\pi}{3}]$$

Q3) Find the value(s) of $x$ (if any ) that satisfy

$$ln(x - 1) = 0$$

Q4) Evaluate the limit

$$\lim_{x\to 0} \frac{\sin(3x^6)}{\tan^2(x^3)}$$

Q5) Evaluate the limit

$$\lim_{x\to 3} \frac{\sqrt{x+1} - 2}{x - 3}$$

Q6) Evaluate the limit

$$\lim_{x\to \infty} \tan^{-1}(\frac{1}{x^2 + 1})$$



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Q7: Study the continuity of the function $f(x) = ln(x^2 + e^x)$

Q8: Study the continuity of the following function at $x = 0$

$$f(x) = \begin{cases} e^{2x} - 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

*Good Luck*

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Q1) $\lim_{x\to 2^+} \frac{1}{4x^2-16} = \frac{1}{0^+} = \infty$

$\lim_{x\to 2^-} \frac{1}{4x^2-16} = \frac{1}{0^-} = -\infty$

$\therefore \lim_{x\to 2} \frac{1}{4x^2-16}$ dn.e

---

Q2) $\sin(3x) = 0$

$3x = 0, \pi, 2\pi$

$x = 0, \frac{\pi}{3}, \cancel{\frac{2\pi}{3}}$

$x = \{0, \frac{\pi}{3}\}$

---

Q3) $\ln(x-1) = 0$

$e^{\ln(x-1)} = e^0 \Rightarrow x-1 = 1$

$x = 1+1 = 2$

$x = 2$

---

Q4) $\lim_{x\to 0} \frac{\sin(3x^6)}{\tan^2(x^3)}$

Sol: $x^6$ على البسط والمقام

$$\lim_{x\to 0} \frac{\frac{\sin(3x^6)}{x^6}}{\frac{\tan^2(x^3)}{x^6}} = \frac{\lim_{x\to 0} \frac{\sin(3x^6)}{x^6}}{\lim_{x\to 0} \frac{\tan(x^3)}{x^3} \cdot \lim_{x\to 0} \frac{\tan(x^3)}{x^3}}$$

$$= \frac{3}{1\cdot 1} = \frac{3}{1} = 3$$

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Q5) $$\lim_{x\to 3} \frac{\sqrt{x+1}-2}{x-3} \cdot \frac{\sqrt{x+1}+2}{\sqrt{x+1}+2}$$

$$= \lim_{x\to 3} \frac{x+1-4}{(x-3)(\sqrt{x+1}+2)} = \lim_{x\to 3} \frac{x-3}{(x-3)(\sqrt{x+1}+2)}$$

$$= \frac{1}{2+2} = \frac{1}{4}.$$

---

Q6) $$\tan^{-1}\left(\lim_{x\to\infty} \frac{1}{x^2+1}\right) = \tan^{-1}\left(\frac{1}{\infty}\right) = \tan^{-1}(0)$$

$$= 0 + n\pi \; ; \; n = 0, \pm 1, \pm 2, \ldots$$

---

Q7) $x^2 + e^x > 0$ always

Therefore $\ln(x^2+e^x)$ cts every where.

---

Q8) $f(0) = 0$

$$\lim_{x\to 0} f(x) \begin{cases} \lim_{x\to 0^+} e^{2x} - 1 = e^0 - 1 = 1 - 1 = 0 \\ \lim_{x\to 0^-} 0 = 0 \end{cases}$$

$\therefore \lim_{x\to 0} f(x) = 0$

$\Rightarrow f(0) = \lim_{x\to 0} f(x)$

$f$ is cts at $x=0$

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1-11-2016

Q1)Circle the correct answer:

1- Given that $f(x) = \frac{x^2+6}{x^2-2x}$ and $g(x) = \frac{1}{x^2-1}$, then $D_{g \circ f}$ is

2- Let $g(x)$ be the function obtained from $f(x) = |x| + x^2$ by shifting 3 units to the right. 2 units up and reflected about the x-axis, then $g(4) =$

a) $-8$ b) $-2$ c) $-14$ d) $-4$

3- If $f(x) = \frac{\cos x}{|x|-4}$, then the domain of $f(x)$ is

a) $\mathbb{R} - 4$ b) $\mathbb{R} - (4,5]$ c) $\mathbb{R} - [4,5)$ d) $\mathbb{R} - [4,5]$

4- $f(x) = ln(1 + lnx)$, then the domain of $f$ is

5- $f(x) = ln(1 + lnx)$, then $f^{-1}(ln2) =$

6- If $f(x) = \frac{4}{4+2\sin^3 x}$, then the range of $f$ is

7- $\cot^2(\tan^{-1} x)$

8- If $f(x) = 2x + 1$ and $f(g(x)) = x^3 + x^2 + x + 1$, then $g(-1) =$

a) 14 b) 7 c) $\frac{-1}{2}$ d) $\frac{1}{2}$



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Q2) Fill The Following Blank:

If $f(x) = \sqrt{(lnx)^2 - 3lnx + 2}$, then the domain of $f$ is



*Good Luck*

*Ahmad Arafeh*

0786543665

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Q1) ① $gof = \dfrac{1}{\left(\frac{x^2+6}{x^2-2x}\right)-1}$

$D\,g(f(x)) = (\mathbb{R} \cap (\mathbb{R}-\{0,2\})) - \{-3\}$

$= (\mathbb{R}-\{0,2\}) - \{-3\}$

$= \mathbb{R} - \{0, 2, -3\}$

أصفار المقام

$\frac{x^2+6}{x^2-2x} - 1 = 0$

$\frac{x^2+6}{x^2-2x} = 1$

$x^2+6 = x^2-2x$

$-2x = 6$

$\Rightarrow x = -3$

---

2) $g(x) \Rightarrow$ 3 unit right  تأثير داخلي

2 unit up  تأثير خارجي

reflection x-axis  $-f(x)$

$g(x) = -(|x-3| + (x-3)^2 + 2)$

$g(4) = -(1+1+2) = -4$

---

3) $[x] - 4 = 0$

$[x] = 4 \quad , \quad 4 \le x < 5$

$Df(x) = \mathbb{R} - [4, 5)$

---

4) $1 + \ln x > 0$

$\ln x > -1$

$e^{\ln x} > e^{-1}$

$x > e^{-1}$

$Df(x) = (e^{-1}, \infty)$

5) $\ln(1+\ln x) = \ln 2$

$e^{\ln(1+\ln x)} = e^{\ln 2}$

$1 + \ln x = 2$

$\ln x = 1 \Rightarrow e^{\ln x} = e^1$

$x = 1$

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6) $-1 \leq \sin^3 x \leq 1$

$-2 \leq 2\sin^3 x \leq 2$

$2 \leq 4 + 2\sin^3 x \leq 6$

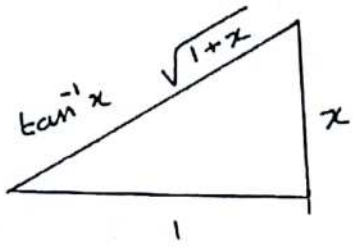
$\frac{1}{6} \leq \frac{1}{4+2\sin^3 x} \leq \frac{1}{2}$

$\frac{4}{6} \leq \frac{4}{4+2\sin^3 x} \leq \frac{4}{2}$

Range $= [\frac{2}{3}, 2]$

7) $(\cot(\tan^{-1} x))^2$

$= \left(\frac{1}{x}\right)^2 = \frac{1}{x^2}$




8) $f(g(-1)) = -1+1-1+1 = 0$

$f(g(-1)) = 0 \Rightarrow f(x) = 2x+1$

$f(g(-1)) = 2\,g(-1)+1$

$0 = 2\,g(-1)+1 \Rightarrow 2\,g(-1) = -1$

$g(-1) = -\frac{1}{2}$

OR

$f(x) = 2x+1$

$f^{-1}(x) = \frac{x-1}{2}$

$y = 2x+1$

$x = 2y+1$

$\frac{x-1}{2} = y$

$f^{-1}(x) = \frac{x-1}{2}$

$f(g(x)) = x^3 + x^2 + x + 1$

$f^{-1}(f(g(x))) = f^{-1}(x^3+x^2+x+1)$

$g(x) = \frac{x^3+x^2+x}{2} \Rightarrow g(-1) = \frac{-1+1-1}{2} = \frac{-1}{2}$

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Q1) ① $gof = \dfrac{1}{\left(\frac{x^2+6}{x^2-2x}\right)-1}$

$D\, g(f(x)) = (\mathbb{R} \cap (\mathbb{R}-\{0,2\})) - \{-3\}$

$= (\mathbb{R}-\{0,2\}) - \{-3\}$

$= \mathbb{R} - \{0,2,-3\}$

أصفار المقام

$\frac{x^2+6}{x^2-2x} - 1 = 0$

$\frac{x^2+6}{x^2-2x} = 1$

$x^2+6 = x^2-2x$

$-2x = 6$

$\Rightarrow x = -3$

---

2) $g(x) \Rightarrow$ 3 unit right  تأثير داخلي

2 unit up  تأثير خارجي

reflection x-axis  $-f(x)$

$g(x) = -(|x-3| + (x-3)^2 + 2)$

$g(4) = -(1+1+2) = -4$

---

3) $[x] - 4 = 0$

$[x] = 4 \quad , \quad 4 \le x < 5$

$Df(x) = \mathbb{R} - [4,5)$

---

4) $1 + \ln x > 0$

$\ln x > -1$

$e^{\ln x} > e^{-1}$

$x > e^{-1}$

$Df(x) = (e^{-1}, \infty)$

5) $\ln(1+\ln x) = \ln 2$

$e^{\ln(1+\ln x)} = e^{\ln 2}$

$1 + \ln x = 2$

$\ln x = 1 \Rightarrow e^{\ln x} = e^1$

$x = 1$

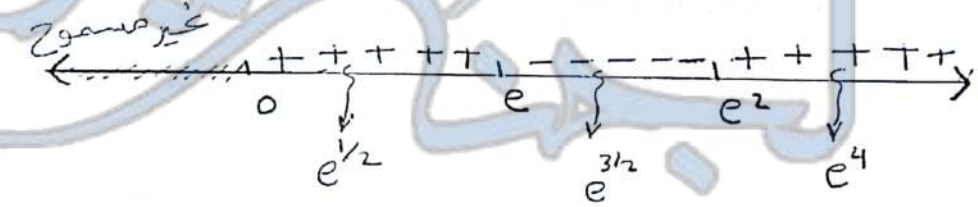
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Q2) $\sqrt{(\ln x)^2 - 3\ln x + 2} = \sqrt{(\ln x - 2)(\ln x - 1)}$

$\ln x = 2 \longrightarrow x = e^2$

$\ln x = 1 \longrightarrow x = e$




D $f(x) = (0, e] \cup [e^2, \infty)$

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Q1) Find the domain of $f(x) = ln(x+1) + \sqrt{4-x^2}$

Q2) The domain of $f(x) = \sin^{-1}(lnx)$ ⟶ $-1 \le \ln x \le 1$

$e^{-1} \le e^{\ln x} \le e^{1}$

$e^{-1} \le x \le e$ $\quad [e^{-1}, e]$

Q3) $\lim_{x\to 0} \frac{\sin^2(3x)}{3x^2} = \lim_{x\to 0} \frac{\sin(3x)}{3x} \cdot \lim_{x\to 0} \frac{\sin(3x)}{x}$

$= 1 \cdot 3 = 3$

Q4) $\lim_{x\to\infty} \sqrt{4x^2+x} - 2x$

Q5) The limit $\lim_{x\to 0^-} \frac{x^3-1}{x^4-x^3} = \frac{-1}{0^+ + 0^+} = \frac{-1}{0^+} = -\infty$

Q6) The number of discontinuities of the function

$$f(x) = \frac{13-x}{5x^2+3\sin^6 x}$$

equals to

Q7) "Final 2017" $f(x) = \frac{x+1}{x-1}$, $x > 1$, then $f^{-1}(x) =$

Q8) "Final 2017" $\lim_{x\to 2} \frac{\sqrt{x^2+12}-4}{x-2}$

Q9) "Final 2017" $\lim_{x\to -1} \frac{\tan(x+1)}{x^2-1} = \frac{\lim_{x\to -1} \frac{\tan(x+1)}{x+1}}{\lim_{x\to -1} \frac{x^2-1}{x+1}}$

Q10) "Final 2017" The domain of function $f(x) = \cos\left(\frac{1}{\sqrt{4-x^2}}\right)$ is

Q11) "Final 2017" The range of the function $f(x) = 2 - \sqrt{x-2}$

Q12) "Final 2017" If $ln(ln(x-1)) = ln2 - ln3$, then $x =$

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Q13) The horizontal asymptote(s) of $f(x) = \frac{\sqrt{x^4+1}}{x-1}$ are

Q14) Mid 2017 $\lim_{x\to 0} \frac{\sin^9(4x)}{x^8} = \lim_{x\to 0} \frac{\sin(4x)}{x} \cdot \frac{\sin(4x)}{x} \cdot \frac{\sin(4x)}{x} \cdots \frac{\sin(4x)}{x} \cdot \sin(4x)$

$= 1 \cdot 1 \cdot 1 \cdot \cdots 1 \cdot 0$

$= 0$

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## أسئلة سنوات على البراباتا (بوليتكنك)

Q1) The limit

$$\lim_{x\to\infty}\left[\sqrt{x^2+(a+2)x}-\sqrt{x^2+2x}\right]$$

$$=\frac{a^a}{2}$$

Then the value of the constant a equals to

Sol:

$$\lim_{x\to\infty}\left[\sqrt{x^2+(a+2)x}-\sqrt{x^2+2x}\right]\cdot\frac{\sqrt{x^2+(a+2)x}+\sqrt{x^2+2x}}{\sqrt{x^2+(a+2)x}+\sqrt{x^2+2x}}$$

$$=\lim_{x\to\infty}\frac{x^2+ax+2x-x^2-2x}{\sqrt{x^2+(a+2)x}+\sqrt{x^2+2x}}$$

$$=\lim_{x\to\infty}\frac{ax}{\sqrt{x^2+(a+2)x}+\sqrt{x^2+2x}}$$

$$=\lim_{x\to\infty}\frac{ax}{|x|+|x|}=\lim_{x\to\infty}\frac{ax}{2x}$$

$$=\frac{a}{2}$$

$$\Rightarrow \frac{a}{2}=\frac{a^a}{2}$$

$$\Rightarrow a^1=a^a \Rightarrow \boxed{a=1}$$

Q2) The H.A for $f(x)=e^{2-x^2}$ is (are)

Sol:

$$\lim_{x\to\infty}e^{2-x^2}=e^{2-\infty}=e^{-\infty}=0$$

$$\lim_{x\to-\infty}e^{2-x^2}=e^{2-(-\infty)^2}=e^{2-\infty}=e^{-\infty}=0$$

$\therefore y=0$ is H.A

Q3) $\lim_{x\to\infty}\frac{2x+1}{\sqrt{3x^2+5}-7}$

Sol: $\lim_{x\to\infty}\frac{2x+1}{\sqrt{3x^2+5}-7}=\frac{\infty}{\infty}$ |

$$\lim_{x\to\infty}\frac{2x}{\sqrt{3}\,|x|}=\lim_{x\to\infty}\frac{2x}{\sqrt{3}\,x}=\frac{2}{\sqrt{3}}$$

Q4) $\lim_{x\to0}\frac{\sin(2x)}{x+\tan(3x)}$

Sol: $\lim_{x\to0}\frac{\sin(2x)}{x+\tan(3x)}=\frac{0}{0}$ |

بقسمة كل من البسط والمقام على x

$$\Rightarrow\lim_{x\to0}\frac{\frac{\sin(2x)}{x}}{\frac{x}{x}+\frac{\tan(3x)}{x}}$$

$$=\frac{2}{1+3}=\frac{2}{4}=\frac{1}{2}$$

Q5) $\lim_{x\to-\infty}\frac{3x-5x^3+1}{x-x^5+1}$

$$\lim_{x\to-\infty}\frac{-5x^3}{-x^5}=\lim_{x\to-\infty}\frac{5}{x^2}$$

$$=\frac{5}{\infty}=0$$

Q6) The limit

$$\lim_{x\to0}x^2\left[\sin^2\left(\cos\left(\frac{5}{x}\right)\right)+\frac{2}{\sin^2(2x)}\right]$$



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Sol :-

$$\lim_{x\to 0} x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right) + \frac{2x^2}{\sin^2(2x)}$$

(I) (II)

(I) $\lim_{x\to 0} x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right)$

$0 \le \sin^2\left(\cos\left(\frac{5}{x}\right)\right) \le 1$

$0 \le x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right) \le x^2$

$0 \to 0 \quad 0$

$0$

(II) $\lim_{x\to 0} \frac{2x^2}{\sin^2(2x)}$

$= 2\left(\lim_{x\to 0} \frac{x}{\sin(2x)}\right)^2$

$= 2 \cdot \left(\frac{1}{2}\right)^2 = \frac{2}{4} = \frac{1}{2}$

$\therefore \lim_{x\to 0} x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right) + \frac{2x^2}{\sin^2(2x)}$

$= 0 + \frac{1}{2} = \frac{1}{2}$

Q7) The V.A of the function $f(x) = \frac{2}{x} + \frac{1}{x^2}$ is

Sol :- $(x=0)$

$\lim_{x\to 0^+} \frac{2}{x} + \frac{1}{x^2} = \frac{2}{0^+} + \frac{1}{0^+}$

$= \infty + \infty = \infty$

$\therefore x=0$ is V.A

Q8) The number of H.A for the function $f(x) = \sqrt{x^2-x} - x^7$ is ?

Sol :-

$\lim_{x\to -\infty} \sqrt{x^2-x} - x^7 =$

$\sqrt{\infty+\infty} - (-\infty)^7 =$

$\infty + \infty = \infty$ $(\frac{N_0}{H.A})$

$\lim_{x\to \infty} \sqrt{x^2-x} - x^7 = \infty - \infty$ !

$\lim_{x\to\infty} \sqrt{x^2-x} - x^7 \cdot \frac{\sqrt{x^2-x}+x^7}{\sqrt{x^2-x}+x^7}$

$= \lim_{x\to\infty} \frac{x^2 - x - x^{14}}{\sqrt{x^2-x}+x^7}$

$= \lim_{x\to\infty} \frac{-x^{14}}{x^7} = \lim_{x\to\infty} -x^7$

$= -\infty$ $(\frac{N_0}{H.A})$

The number of H.A(s) is 0.

Q9) $\lim_{x\to 1} \frac{e^{2\ln x} - 9}{x^3 - 3x^2 + x - 3}$

Sol :- $= \frac{e^{2\ln 1} - 9}{1-3+1-3}$

$= \frac{e^0 - 9}{-4} = \frac{1-9}{-4} = \frac{-8}{-4} = 2$

Q10) $\lim_{x\to 2^-} \left(\frac{x+2}{2-x}\right) = \frac{4}{0^+} = \infty$



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Q11) $\lim_{x\to\infty} \sqrt{x^2-5} - x = \infty - \infty$ !

Sol :- $\lim_{x\to\infty} \frac{\sqrt{x^2-5}-x \cdot \sqrt{x^2-5}+x}{\sqrt{x^2-5}+x}$

$= \lim_{x\to\infty} \frac{x^2-5-x^2}{\sqrt{x^2-5}+x}$

$= \lim_{x\to\infty} \frac{-5}{\sqrt{x^2-5}+x} =$

$= \frac{-5}{\infty} = 0$

Q12) $\lim_{x\to\infty} \sqrt{x^2+4x} - \sqrt{x^2+x}$

Sol :- $= \infty - \infty$ !

$\lim_{x\to\infty} \sqrt{x^2+4x} - \sqrt{x^2+x} \cdot \frac{\sqrt{x^2+4x}+\sqrt{x^2+x}}{\sqrt{x^2+4x}+\sqrt{x^2+x}}$

$= \lim_{x\to\infty} \frac{x^2+4x-x^2-x}{\sqrt{x^2+4x}+\sqrt{x^2+x}}$

$= \lim_{x\to\infty} \frac{3x}{|x|+|x|} = \lim_{x\to\infty} \frac{3x}{2x}$

$= \frac{3}{2}$ .

Q13) If $f(x) = \frac{x^2-1}{x^2-2x+1}$ , then

H.A and V.A are :-

Sol :- H.A $\lim_{x\to\pm\infty} \frac{x^2-1}{x^2-2x+1}$

$= \lim_{x\to\pm\infty} \frac{x^2}{x^2} = 1$

∴ $y=1$ is H.A

V.A $x=1$ . أصفار المقام

$\lim_{x\to1} \frac{x^2-1}{x^2-2x+1} = \lim_{x\to1} \frac{(x-1)(x+1)}{(x-1)(x-1)}$

$= \frac{2}{0} = \pm\infty$

∴ $x=1$ is V.A

Q14) $\lim_{x\to\infty} \sqrt{x+3\sqrt{x}} - \sqrt{x}$

Sol :- $= \infty - \infty$ !

$\Rightarrow \lim_{x\to\infty} \sqrt{x+3\sqrt{x}} - \sqrt{x} \cdot \frac{\sqrt{x+3\sqrt{x}}+\sqrt{x}}{\sqrt{x+3\sqrt{x}}+\sqrt{x}}$

$= \lim_{x\to\infty} \frac{x+3\sqrt{x}-x}{\sqrt{x+3\sqrt{x}}+\sqrt{x}}$

$= \lim_{x\to\infty} \frac{3\sqrt{x}}{\sqrt{x}\left(\sqrt{1+\frac{3}{\sqrt{x}}}+1\right)}$

$= \frac{3}{\sqrt{1+0}+1} = 3/2$

Q15) If $\lim_{x\to0} \frac{\sqrt{ax+b}-2}{x} = 4$

then the values of a and b are :

Sol :- إذا كانت النهاية موجودة و
نهاية المقام = صفر فإن
نهاية البسط = صفر

$\lim_{x\to0} x = 0 \Rightarrow \lim_{x\to\infty} \sqrt{ax+b}-2 = 0$

$\sqrt{b} = 2$

$\Rightarrow b = 4$



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$$\lim_{x\to 0} \frac{\sqrt{ax+4}-2}{x} \cdot \frac{\sqrt{ax+4}+2}{\sqrt{ax+4}+2}$$

$$= \lim_{x\to 0} \frac{ax+4-4}{x(\sqrt{ax+4}+2)}$$

$$= \lim_{x\to 0} \frac{a}{\sqrt{ax+4}+2} = \frac{a}{4}$$

$$\Rightarrow \frac{a}{4} = 4$$

$$\Rightarrow a = 16$$

Q16.) If $\lim_{x\to a}[f(x)+g(x)] = 3$ and $\lim_{x\to a}[f(x)-g(x)] = 2$ then $\lim_{x\to a} \frac{f(x)}{g(x)} =$

Sol:-

(I) $\lim_{x\to a}[f(x)+g(x)] +$

$\lim_{x\to a}[f(x)-g(x)] = 3+2$

$\Rightarrow \lim_{x\to a}[f(x)+g(x)+f(x)-g(x)] = 5$

$\Rightarrow 2\lim_{x\to a} f(x) = 5$

$\Rightarrow \lim_{x\to a} f(x) = \frac{5}{2}$

(II) $\lim_{x\to a}[f(x)+g(x)] - \lim[f(x)-g(x)]$

$= 3-2$

$\Rightarrow \lim_{x\to a}[f(x)+g(x)-f(x)+g(x)]$

$= 1$

$\Rightarrow 2\lim_{x\to a} g(x) = 1$

$\lim_{x\to a} g(x) = 1/2$.

$\therefore \lim_{x\to a} \frac{f(x)}{g(x)} = \frac{5/2}{1/2} = \frac{5}{1} = 5$.

Q17.) $\lim_{x\to\infty} \frac{3x^2+\cos^2 x}{x^2+1}$

Sol:- $\lim_{x\to\infty} \frac{\frac{3x^2}{x^2}+\frac{\cos^2 x}{x^2}}{\frac{x^2}{x^2}+\frac{1}{x^2}}$

$= \frac{3+0}{1+0} = 3$

$+ \lim_{x\to\infty} \frac{\cos^2 x}{x^2}$

$0 \le \cos^2 x \le 1$

$0 \le \frac{\cos^2 x}{x^2} \le \frac{1}{x^2}$

$0 \quad\quad 0$

$0$

Q18.) The H.A(s) of $f(x) = \sqrt{x^2+x+1} - \sqrt{x^2-4x}$ are (H.W)

Sol:-

$y = \frac{-5}{2}, \frac{5}{2}$ are H.A(s)



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Q19) Let $f(x) = \frac{5}{x^2 - Kx + \frac{9}{4}}$ then the values of K such that f has no vertical ?

Sol:- no vertical = لا أصفار للمقام

∴ لا يكون للمعادلة التربيعية أصفار إذا كان مميزها أقل من صفر

$b^2 - 4ac < 0$

$K^2 - 4(1)(\frac{9}{4}) < 0$

$K^2 - 9 < 0 \Rightarrow K^2 < 9$

$\Rightarrow \sqrt{K^2} < \sqrt{9}$

$\Rightarrow |K| < 3$

$-3 < K < 3$

$K \in (-3, 3)$

Q20) $\lim_{x \to \infty} \frac{x + \sin 5x}{5x + \sin x}$

Sol:- x على البسط والمقام

$$\lim_{x \to \infty} \frac{\frac{x}{x} + \frac{\sin 5x}{x}}{\frac{5x}{x} + \frac{\sin x}{x}}$$

$$= \lim_{x \to \infty} \frac{1 + \frac{\sin 5x}{x}}{5 + \frac{\sin x}{x}} = \frac{1+0}{5+0}$$

$$= \frac{1}{5}$$

$-1 \le \sin x \le 1$ , $-1 \le \sin 5x \le 1$

$\frac{-1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$ , $\frac{-1}{x} \le \frac{\sin 5x}{x} \le \frac{1}{x}$

$0 \le 0 \le 0$ , $0 \le 0 \le 0$

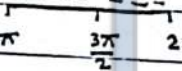
Q21) The vertical asy. of $f(x) = \ln(x-2)$ is

Sol:- $x = 2$.

$\lim_{x \to 2^+} \ln(x-2) = \ln 0^+ = -\infty$

$x = 2$ V.A

Q22) $\lim_{x \to \frac{3\pi}{2}^+} \frac{|\sin x|}{\sin x}$




Sol:- $\lim_{x \to \frac{3\pi}{2}^+} \frac{-\sin x}{\sin x} = -1$

Q23) $\lim_{x \to 2^+} (x-2) \sin\left(\frac{1}{x^2-4}\right)$

Sol:- = 0 (dn.c) !

$-1 \le \sin\left(\frac{1}{x^2-4}\right) \le 1$

$-(x-2) \le (x-2)\sin\left(\frac{1}{x^2-4}\right) \le (x-2)$

$\lim_{x \to 2^+} -(x-2) = 0$ , $\lim_{x \to 2^+} (x-2) = 0$

0



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Q24) If $f(x) = \frac{2x^2+1}{(2x-1)^2-1}$, then H.A and V.A are :- (H.W)

Sol :- H.A $y = \frac{1}{2}$

V.A $x=1$ and $x=0$

Q25) $\lim_{x\to 3} (x^2-9)\cot(x^2-x-6)$

Sol :- $0 \cdot \infty$ !

$$\lim_{x\to 3} (x^2-9)\frac{\cos(x^2-x-6)}{\sin(x^2-x-6)}$$

$$= \lim_{x\to 3} \cos(x^2-x-6) \cdot \lim_{x\to 3} \frac{x^2-9}{\sin(x^2-x-6)}$$

$$= 1 \cdot \lim_{x\to 3} \frac{x^2-9}{\frac{\sin(x^2-x-6)}{x^2-x-6}}$$

$$= \lim_{x\to 3} \frac{(x-3)(x+3)}{(x-3)(x+2)}$$

$$= \frac{5}{6}$$

Q26) $\lim_{x\to -\infty} \sqrt{x^2+4x} - \sqrt{x^2-x}$

Sol :- $\lim_{x\to -\infty} \sqrt{x^2+4x} - \sqrt{x^2-x}$ .

$$\frac{\sqrt{x^2+4x}+\sqrt{x^2-x}}{\sqrt{x^2+4x}+\sqrt{x^2-x}}$$

$$= \lim_{x\to -\infty} \frac{5x}{\sqrt{x^2+4x}+\sqrt{x^2+x}}$$

$$= \lim_{x\to -\infty} \frac{5x}{|x|+|x|}$$

$x \to -\infty$ , $|x| \to -x$

$$= \lim_{x\to -\infty} \frac{5x}{-2x} = \frac{-5}{2}$$

Q27) If $\lim_{x\to 1} \frac{x^3+2x^2-x+a}{x^2-1}$ exists, then $a = ??$

Sol :- بما أن النهاية موجودة ونهاية المقام = صفر ⇐ نهاية البسط = صفر

$$\Rightarrow \lim_{x\to 1} x^3+2x^2-x+a = 0$$

$$1+2-1+a = 0$$

$\therefore a = -2$

Q28) a) $\lim_{x\to 0} \frac{\sqrt{x+9}-3}{x}$ (H.W)

Sol :- $\frac{1}{6}$

b) $\lim_{x\to 0^-} 3x \sin(\frac{2}{x})$ [H.W]

Sol :- 0



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Q29) H.A for

$$f(x)=\frac{2x^2+1}{x+2x^2}+\frac{x-2}{3-|x|}$$

Sol:

$$\lim_{x\to\infty}\frac{2x^2+1}{x+2x^2}+\frac{x-2}{3-|x|}$$

$$=\lim_{x\to\infty}\frac{2x^2}{2x^2}+\frac{x}{-x}$$

$$=1-1=0$$

$\therefore y=0$ H.A

$$\lim_{x\to-\infty}\frac{2x^2}{2x^2}+\frac{x}{x}$$

$$=1+1=2$$

$\therefore y=2$ H.A

$y=0, 2$ are H.A

Q30) $\lim_{x\to3^-}\frac{x^2-3x}{x^2-6x+9}$

Sol:

$$=\frac{0}{0}\ !$$

$$\lim_{x\to3^-}\frac{x(x-3)}{(x-3)(x-3)}=\frac{3}{0^-}$$

$$=-\infty$$

Q31) $\lim_{x\to3}\frac{\sin(x-3)}{x^2-2x-3}$

Sol:

$$=\frac{0}{0}\ !$$

$$\lim_{x\to3}\frac{\sin(x-3)}{(x-3)(x+1)}$$

$$=\lim_{x\to3}\frac{\sin(x-3)}{x-3}\cdot\lim_{x\to3}\frac{1}{x+1}$$

$$=1\cdot\frac{1}{4}=\frac{1}{4}$$

Q32) 6-12-2016

$$\lim_{x\to\infty}\sqrt{ax^2+bx}-3x=2$$

then the values of a and b are :-

Sol:

$$\lim_{x\to\infty}\sqrt{ax^2+bx}-3x\cdot\frac{\sqrt{ax^2+bx}+3x}{\sqrt{ax^2+bx}+3x}=2$$

$$\Rightarrow\lim_{x\to\infty}\frac{ax^2+bx-9x^2}{\sqrt{ax^2+bx}+3x}=2$$

$$\Rightarrow\lim_{x\to\infty}\frac{(a-9)x^2+bx}{\sqrt{ax^2+bx}+3x}=2$$

بما أن النهاية موجودة وتساوي عدد فإنه يجب أن تكون أعلى قوة في البسط = أعلى قوة في المقام . لهذا يجب أن يكون معامل $x^2$ في البسط = صفر

$$\Rightarrow a=9$$

$$\Rightarrow\lim_{x\to\infty}\frac{bx}{3x+3x}=2$$

$$\lim_{x\to\infty}\frac{bx}{6x}=2$$

$$\frac{b}{6}=2\Rightarrow b=12$$



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Q33) 2016 $\lim_{x\to 1} \frac{(x-1)}{\sqrt{x^2-2x+1}}$

Sol: $\frac{0}{0}$ !

$$\lim_{x\to 1} \frac{x-1}{\sqrt{(x-1)^2}} = \lim_{x\to 1} \frac{(x-1)}{|x-1|}$$

$|x-1|$: − − − − | + + + +  (1)

$\lim_{x\to 1^+} \frac{x-1}{x-1} = 1$

$\lim_{x\to 1^-} \frac{x-1}{-(x-1)} = -1$ ⟹

$\lim_{x\to 1} \frac{x-1}{|x-1|}$ d.n.e.

Q34) 2016 $\lim_{x\to\infty} 3x \tan\left(\frac{2}{x}\right)$

Sol: $\infty . 0$ ! استبدال

$y = \frac{1}{x} \Rightarrow x = \frac{1}{y}$

$x \to \infty$ , $y \to 0$

$$\lim_{y\to 0} \frac{3\tan(2y)}{y}$$

$$= 3 \lim_{y\to 0} \frac{\tan(2y)}{y}$$

$$= 3 \cdot \frac{2}{1} = 6$$

Q35) 2016 The H.A and V.A for $f(x) = e^{\left(\frac{x+\sin 3x}{2x}\right)}$

Sol: V.A $x = 0$

$\lim_{x\to 0} e^{\left(\frac{x+\sin 3x}{2x}\right)}$

$= e^{\lim_{x\to 0}\left(\frac{x}{2x} + \frac{\sin 3x}{2x}\right)}$

$= e^{\frac{1}{2}+\frac{3}{2}} = e^{2} \neq \pm\infty$

∴ $x = 0$ (Not V.A)

$\lim_{x\to\pm\infty} e^{\left(\frac{x+\sin 3x}{2x}\right)}$

$= e^{\lim_{x\to\pm\infty}\left(\frac{x}{2x} + \frac{\sin 3x}{2x}\right)}$ ← Squezing

$= e^{\frac{1}{2}+0}$

$= \sqrt{e}$

∴ $y = \sqrt{e}$ ... H.A

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* حل أسئلة سنوات

Q) $\lim_{x \to 0} \cos\left(\frac{\sin(x+\pi)}{2x}\right)$

Sol. $\cos\left(\lim_{x \to 0} \frac{\sin(x+\pi)}{2x}\right)$

$\frac{0}{0}$

$\cos\left(\lim_{x \to 0} \frac{\sin x \cos\pi + \sin\pi \cos x}{2x}\right)$

$\cos\left(\lim_{x \to 0} \frac{-\sin x}{2x}\right) = \cos\left(\frac{-1}{2}\right)$

$= \cos\left(\frac{1}{2}\right)$



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Q) $\lim_{x\to\infty} [\ln(3e^x - e^{-x}) - \ln(e^x - e^{-x})]$

Sol. $\infty - \infty$

$$\lim_{x\to\infty} \ln\left(\frac{3e^x - e^{-x}}{e^x - e^{-x}}\right)$$

$$\ln\left(\lim_{x\to\infty} \frac{3e^x - e^{-x}}{e^x - e^{-x}}\right)$$

$$= \ln\left(\lim_{x\to\infty} \frac{e^x(3 - e^{-2x})}{e^x(1 - e^{-2x})}\right)$$

$$= \ln\left(\frac{3}{1}\right) = \ln 3$$

2) The Continuity Points for

$f(x) = \dfrac{\ln(x-1)}{x^2+x+1}$ are

Sol. $\ln(x-1)$ is cts on $(1,\infty)$

$x^2+x+1$ is cts on $\mathbb{R}$

$b^2 - 4ac = -3 < 0$ ليس للمقام أصفار

$f(x)$ cts on $(1,\infty) \cap \mathbb{R} = (1,\infty)$

Q) $\lim_{x\to-\infty} \cos^{-1}\left(\frac{1-x^2}{2x^2+1}\right)$

Sol. $\cos^{-1}\left(\lim_{x\to-\infty} \frac{-x^2}{2x^2}\right)$

$$= \cos^{-1}\left(\frac{-1}{2}\right) = \frac{2\pi}{3}$$

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Q) The limit
$\lim_{x\to 2} 3\sin^{-1}\left(\frac{x^2-5x+6}{x^2-2x}\right)$

Sol.
$3\sin^{-1}\left(\lim_{x\to 2}\frac{x^2-5x+6}{x^2-2x}\right)$

$3\sin^{-1}\left(\lim_{x\to 2}\frac{(x-3)(x-2)}{x(x-2)}\right)$

$= 3\sin^{-1}\left(\frac{-1}{2}\right) = -3\sin^{-1}\left(\frac{1}{2}\right)$

$= -3\cdot\frac{\pi}{6} = \frac{-\pi}{2}$

Q) H.W $f(x) = \begin{cases} \frac{\tan kx}{x}, & x<0 \\ 2x+3k^2, & x\geq 0 \end{cases}$

all values of k that let $\lim_{x\to 0} f(x)$ exists are

Sol. $k = 0, \frac{1}{3}$

2) $\lim_{x\to 0} \tan\left(\frac{1-\cos x}{x}\right)$

Sol. $\tan\left(\lim_{x\to 0}\frac{1-\cos x}{x}\right)$

$\tan\left(\lim_{x\to 0}\frac{1-\cos x}{x}\cdot\frac{1+\cos x}{1+\cos x}\right)$

$= \tan\left(\lim_{x\to 0}\frac{1-\cos^2 x}{2x}\right)$

$= \tan\left(\lim_{x\to 0}\frac{\sin^2 x}{2x}\right)$

$= \tan\left(\lim_{x\to 0}\frac{\sin x}{2x}\cdot\lim_{x\to 0}\sin x\right)$

$= \tan(0) = 0$

Q) Let $f(x) = \sqrt{|x|-4}$, then f Continuous on

Sol. $|x|-4\geq 0$

$|x|\geq 4 \Rightarrow x\geq 4$ or $x\leq -4$

$[4,\infty)$, $(-\infty,-4]$

Q) All values of a such that

$f(x) = \begin{cases} x+4, & x\leq a \\ x^2, & x>a \end{cases}$

cts on $\mathbb{R}$ are

Sol. $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$

$\lim_{x\to a^+} x^2 = \lim_{x\to a^-} x+4$

$a^2 = a+4$

$a^2 - a - 4 = 0$

$a = \frac{1\pm\sqrt{1-4(1)(-4)}}{2(1)} = \frac{1\pm\sqrt{17}}{2}$

Q) $f(x) = \frac{\csc x}{|x-1|-2}$ is discontinuous at

Sol.
$f(x) = \frac{1}{\sin x(|x-1|-2)}$

$\sin x = 0$, $x = \pm n\pi$ أصفار المقام

$|x-1|-2=0 \rightarrow x-1=2 \Rightarrow x=3$
$\rightarrow x-1=-2 \Rightarrow x=-1$

discts at $x = \pm n\pi, -1, 3$



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# أسئلة سنوات

## مرفق معها الإجابة النهائية



(اختبر نفسك )

٦١ سؤال

Chapter zero

أحمد عرفه

٠٧٨٦٥٤٣٦٦٥

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( Calculus ) المادة : إعداد الاستاذ : أحمد عرفه

Q1: Simplify the expression
$\cos(2\tan^{-1}x)$

Q2: Find $(f\circ g)(x)$ if
$f(x)=x^2-1$, $g(x)=2x+1$

Sol $f(g(x))=4x^2+4x$

Q3: The domain of
$F(x)=\sqrt{4-x^2}+\frac{1}{x-2}$ is?

Sol: $[-2,2)$

Q4: Find $(f\circ g\circ h)(x)$:
$f(x)=x+1$, $g(x)=2x$,
$h(x)=x-1$

Sol $-1+2x$

Q5: The domain of
$h(x)=\sqrt{4-x^2}\ \sin^{-1}(2x-4)$
is:

Sol $[\frac{3}{2},2]$

Q6: If $f(x)=\frac{\cos^{-1}(1-x)}{2-x}$,
then the domain is:

Sol: $[0,2)$.

Q7: let $(x)$ be a number in the interval $[-1,1]$ then $\csc(\sin^{-1}x)$ is

Sol $\frac{1}{x}$

Q8: If $f(x)=\begin{cases}2x^2+4, & x\le 1\\ x-1, & 1<x\le 4\\ x+1, & x>4\end{cases}$
then $f\circ f(1)$ is

Sol 7

Q9: If $\frac{3\pi}{2}<\theta<2\pi$ and
$\cos\theta=\frac{1}{4}$, then $\csc\theta=?$

Sol $\frac{-4}{\sqrt{15}}$

Q10: Find the Range of
$f(x)=5\sin^2x+2$ is

Sol $[2,7]$

Q11: The inverse function of
$f(x)=\frac{1}{3+3\tan^{-1}x}$

Sol: $f^{-1}(x)=\tan\left(\frac{1-3x}{3x}\right)$

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Q12: The range of $f(x) = \frac{6x-2}{3x+1}$ is

Sol $\mathbb{R} - \{2\}$.

Q13: which of the following an even function?

a) $x^2+3x$ b) $2-3x$
c) $1+2\cos x$ d) $\frac{x^3-x}{1+x^2}$

Q14: The domain of $f(x) = 2\sin^{-1}(1-2x)$ is

Sol $[0,1]$.

Q15: The domain of $f(x) = \frac{1}{\frac{2}{x-1}-1}$ is

Sol $\mathbb{R} - \{1,3\}$.

Q16: If $f(x) = x^3+2x+5$ then $f^{-1}(8) =$

Sol = 1.

Q17: The range of $g(x) = -x^2+2x-5$ is

Sol $(-\infty, -4]$.

Q18: If $f(x) = 3x-5$ and $(f \circ g)(x) = 12x^2-6x+13$, then $g(0) = ?$

Sol 6

Q19: If $f(x) = \frac{5}{x-3}$ and $g(x) = x^2-1$, then the domain of $(f \circ g)(x)$ is

Sol $\mathbb{R} - \{-2, 2\}$.

Q20: The function $g(x)$ is obtained from $f(x) = x^2-2x$ by shifting $f(x)$ one unit to the left then reflecting about the y-axis, then $g(x)$ is?

Sol $g(x) = x^2-1$

Q21) $\sin(\tan^{-1}2)$

Sol: $\frac{2}{\sqrt{5}}$

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إعداد الاستاذ : أحمد عرفه — المادة : (Calculus 1)

Q22) The domain of $f(x)=4\sin^{-1}(4x+1)$ is

a) $[-0.5,0]$ b) $[-1,1]$
c) $[-1,0]$ d) $[-2,2)$

Q23) Let $f(x)=3+7\sqrt{4-x}$, If $g(x)$ is a function such that the domain of $g$ is $(0,\infty)$, then the domain of multiplication of the functions $f.g(x)$ is

a) $(0,4)$ b) $(0,4]$
c) $(0,\infty)$ d) $(4,\infty)$

Q24) Assume $g(x)=x^2+5$, Range $f(x)=[2,5]$ Find Range $gof(x)$ ?

a) $[9,30]$ b) $(9,30]$
c) $(9,30)$ d) $[2,5]$

Q25) Let $f(x)=\frac{3}{x-3}$ and $g(x)=\frac{2}{x}$, then $D_{gof(x)}$ ?

Sol. $\mathbb{R}-\{3\}$

Q26) The domain of $f(x)=\frac{\cos x}{1-\sin x}$ is

Sol. $\mathbb{R}-\{\frac{\pi}{2}+2n\pi,\ n=0,1,2,\dots\}$

Q27) The inverse function of $f(x)=8+\sqrt[3]{x-4}$

Sol. $(x-8)^3+4$

Q28) $\cot(\sin^{-1}\frac{3}{7})$

Sol. $\frac{2\sqrt{10}}{3}$

Q29) $\cos(\text{arc}\sin(\frac{-1}{4}))$ is

Sol. $\frac{\sqrt{15}}{4}$

Hint: $\text{arc}\sin x=\sin^{-1}x$

Q30) The range of $g(x)$ is $[-5,5]$ and $f(x)=g^2(x)$, then the range of $f(x)$ is

Sol. $[0,25]$

Q31) The range of $f(x)=2+\sqrt{x-1}$

a) $(2,\infty)$ b) $[2,\infty)$
c) $(-\infty,2)$ d) $(-\infty,-2]$

Q32) The domain of $f(x)=\sqrt{2-|x+3|}$ is

Sol. $[-5,-1]$

Q33) which of the following function is Symmetric about the origin?

a) $|x|+\sin x$ b) $x\sin x$
c) $x\cos x$ d) $|x|$

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إعداد الاستاذ : أحمد عرفه — المادة : (Calculus 1)

Q34) If $f(x)=\frac{1+x}{1-x}$ and $g(x)=\frac{x}{1-x}$
Find $D_{fog}(x)$?
Sol. $\mathbb{R}-\{\frac{1}{2},1\}$

Q35) The equation of $\tan y = 1+x^2$ Symmetric about :
Sol. y-axis only

Q36) If the domain of $f(x)$ is $[-5,5]$ and the range is $[1,4]$, then the domain of $3\sqrt{f(2x-3)}+1$ is :
Sol. $[-1,4]$

Q37) If $f(x)=\sqrt{x}$, $g(x)=x^3+1$, Find $g(f(4))$?
Sol. 9

Q38) The domain of $f(x)=\frac{1}{(x-3)^5}$
Sol. $\mathbb{R}-\{3\}$

Q39) Find the value of $fogoh(x)$ If $f(x)=\sqrt{x}$, $g(x)=\frac{1}{x}$, $h(x)=x^3$
Sol. $x^{-3/2}$

Q40) If $g(x)=\sqrt{x}$, Find $g(\frac{1}{\sqrt{x}})$?
Sol. $x^{-1/4}$

Q41) The value of $\sin(\frac{5\pi}{6})$
Sol. $\frac{1}{2}$

Q42) The value of $\cos(\frac{4\pi}{3})$
Sol. $=\frac{1}{2}$

Q43) If $f(x)=\sin x$, $g(x)=x^2+3$ Find $gof(\pi)$
Sol. 3

Q44) The domain of $f(x)=\sqrt{9-x^2}\cdot\sqrt{x-1}$
Sol. $[1,3]$

Q45) If $f(x)=5x-1$ $g(x)=\frac{x}{2+x}$, Find $f(g(1))$
Sol. $\frac{2}{3}$

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(Calculus) المادة : إعداد الاستاذ : أحمد عرفه

Q46) The domain of $f(x)=\frac{\sqrt{x-2}+4}{x-5}$

Sol: $[2,\infty)-\{5\}$

Q47) If $f(x)=x^3+x+a$, an odd function, Find a?

Sol: $a=0$

Q48) Express the function in the form fogoh(x)
$H(x)=\sqrt[8]{2+|x|}$

Sol: $f(x)=\sqrt[8]{x}$, $g(x)=2+x$, $h(x)=|x|$

Q49) Express the function in the form fogoh(x):
$H(x)=\sec^4\sqrt{x}$

Sol $f(x)=x^4$, $g(x)=\sec x$, $h(x)=\sqrt{x}$

Q50) $\sin(2\cos^{-1}(\frac{3}{5}))$
Sol: $\frac{24}{25}$

Q51) The figure shows the graph of (Sin) function for one complete period, which of the following is an equation for the graph?

a) $y=2\sin(\frac{x}{2})$ b) $y=2\sin x$
c) $y=2\sin(2x)$ d) $y=\sin(\pi x)$

Q52) Find domain gog(x) if $g(x)=\sqrt[3]{1-x}$
Sol: R

Q53) Find the value of $\sin(\cos^{-1}(\frac{3}{5}))+\tan(\cos^{-1}(\frac{3}{5}))$

Sol: $\frac{4}{5}+\frac{4}{3}=\frac{32}{15}$

Q54) Find g(x) If $f(x)=\frac{x}{x+1}$ and $fog(x)=\frac{1}{x}$?

Sol: $g(x)=\frac{1}{x-1}$

Q55) One of the following functions is not one to one:
a) $f(x)=\tan^{-1}x$ b) $f(x)=|x+1|+2$
c) $f(x)=x^3$

Q56) The graph of $y=x^2\tan x$ is Symmetric about Sol origin

Q57) Let $f(x)=\frac{3}{x-1}$ and $g(x)=4x$, then the Solution(s) of $f(g(x))=g(f(x))$ is (are)?

a) 2 b) $\frac{1}{5}$
c) $\frac{1}{5}, 2$ d) 5

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Q58) Is the function $f(x)=\frac{\sin(x^2)}{x^3-2x}$ odd, even or neither?

Sol. odd.

Q59) Find the inverse of f

$$y=-\sqrt{\frac{2}{x}-1}$$

Sol. $f^{-1}(x)=\frac{2}{x^2+1}$

Q60) Find the domain of f

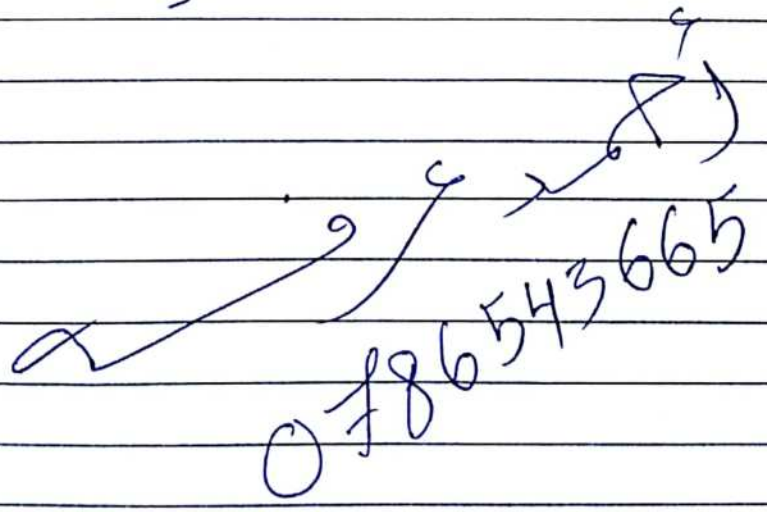
$$f(x)=\frac{5x+4}{|5x+4|}$$

Sol. $(-\infty,\frac{4}{5})\cup(\frac{4}{5},\infty)$

Q61) Find the inverse of

$$f(x)=(3x-2)^5+7$$

Sol $f^{-1}(x)=\frac{\sqrt[5]{x-7}+2}{3}$



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